

CSCI 340 DATABASE DESIGN

Relational Algebra Practice

Selection, projection, joins, and set operations

Name _____ People consulted _____

Purpose. Build fluency with relational algebra by reading expressions, tracing exact results, and writing expressions of increasing complexity. Every question uses the same small registration database.

Directions. Submit your own completed worksheet. You may work with classmates, but list everyone you worked with above. Unless a question says otherwise, give only the requested attributes. When asked to trace an expression, write the output schema and every resulting tuple. Tuple order does not matter because a relation is an unordered set.

Notation and assumptions

Operation	Form	Effect
Selection	$\sigma_p(E)$	Keeps tuples of E that satisfy predicate p .
Projection	$\pi_{A_1, \dots, A_k}(E)$	Keeps the listed attributes and removes duplicate output tuples.
Natural join	$E_1 \bowtie E_2$	Matches on every shared attribute name and keeps one copy of each shared attribute.
Theta join	$E_1 \bowtie_{\theta} E_2$	Matches pairs satisfying θ and keeps attributes from both inputs.
Set operations	$E_1 \cup E_2, E_1 \cap E_2, E_1 - E_2$	Combine union-compatible inputs as sets of tuples.

Use qualified names such as `student.student_id` when an expression could otherwise be ambiguous. You may use assignment notation, $x \leftarrow E$, to name an intermediate result. String values appear in quotation marks.

Shared registration database

Reference for every section

The schema has three relations:

`student(student_id, student_name, major, year)`

`course(course_id, course_title, department, credits)`

`enrollment(student_id, course_id, grade)`

Relation: student

student_id	student_name	major	year
101	Ava	CS	junior
102	Jordan	Math	sophomore
103	Lin	Biology	senior
104	Sam	CS	sophomore
105	Noor	Math	junior
106	Mira	History	first-year

Relation: course

course_id	course_title	department	credits
C10	Databases	CS	3
C20	Discrete Math	Math	3
C30	Ecology	Biology	4
C40	Data Ethics	CS	3
C50	World History	History	4

Relation: enrollment

student_id	course_id	grade
101	C10	A
101	C40	B
102	C20	A
103	C30	B
104	C10	B
105	C10	A

Facts worth noticing

- A student may have more than one enrollment.
- A course may have more than one enrolled student.
- A student's major need not match a course's department.
- A student or course may have no matching enrollment.

Section A	Selection and projection	12 points
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A1 Read expressions in plain English	3 points
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For each expression, state in plain English exactly what its result contains.

- a. $\sigma_{major=\text{"CS"}}(\text{student})$
- b. $\pi_{student_name, year}(\sigma_{major=\text{"Math"}}(\text{student}))$
- c. $\pi_{department}(\text{course})$

A2 Trace exact results	5 points
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Write the output schema and every tuple produced by each expression. Explicitly identify any duplicate tuple removed by projection.

- a. $\pi_{major}(\text{student})$

- b. $\pi_{student_id, student_name}(\sigma_{year=\text{"sophomore"}}(\text{student}))$

A3 Write expressions	4 points
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Write one relational-algebra expression for each request.

- a. Return the title of every 4-credit course. (2)

- b. Return the name and major of every junior student. (2)

Section B**Natural join and theta join 17 points**

B1 Distinguish the joins**3 points**

Answer each question about the expression `student ⋈ enrollment`.

- Which shared attribute is used automatically as the matching condition?
- Why does that shared attribute appear only once in the result schema?
- If the two relations had no shared attribute names, what would their natural join produce?

B2 Trace a natural join**6 points**

Write the output schema and every tuple in `student ⋈ enrollment`. Then name the student who does not appear and explain why. Tuple order does not matter.

B3 Analyze a theta join**4 points**

Consider the expression

`student ⋈student.major=course.department course.`

- State what the expression means in plain English. (1)
- State its output schema. (1)
- How many output tuples are there? Briefly justify your count; do not list every tuple. (2)

B4 Write composed join queries**4 points**

Write one relational-algebra expression for each request. Natural join is appropriate because the linked relations share deliberately matching attribute names.

a. Return the names of students enrolled in course C10. (2)

b. Return the titles of courses taken by Math majors. (2)

Section C**Set operations and composition 16 points**

For C1–C3, use these named intermediate results:

$$\begin{aligned}cs_student &\leftarrow \pi_{student_id}(\sigma_{major=\text{“CS”}}(student)) \\sophomore &\leftarrow \pi_{student_id}(\sigma_{year=\text{“sophomore”}}(student))\end{aligned}$$

C1 Check compatibility**3 points**

Are *cs_student* and *sophomore* union-compatible? State the schema of each result and explain briefly.

C2 Read set expressions in plain English**3 points**

State what each expression means in plain English. Be precise about the direction of difference.

- a. $cs_student \cup sophomore$
- b. $cs_student \cap sophomore$
- c. $cs_student - sophomore$

C3 Trace set results**4 points**

First write every tuple in *cs_student* and *sophomore*. Then write every tuple in the union, intersection, and difference from C2. Explain why a student ID appearing in both inputs occurs only once in the union.

Section C continued**Composition with joins**

C4 Compose set operations with joins**6 points**

First define the IDs of students enrolled in a CS-department course:

$$takes_cs \leftarrow \pi_{student_id} (enrollment \bowtie \sigma_{department="CS"}(course)) .$$

Using $cs_student$ and $takes_cs$, write an expression for each request. You do not need to enumerate the results.

- a. IDs of students who major in CS or take a CS course.

- b. IDs of students who major in CS and take a CS course.

- c. IDs of students who take a CS course but do not major in CS.